Big Data Analytics

::Final project 2026 (50%)

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No.	Question	Grade
1	Problem Definition & Decision Questions (1.0)	
2	Dataset Selection & Data Quality (1.5)	
3	MySQL Database & SQL Queries (1.0)	
4	KNIME Workflow & Data Processing (1.5)	
5	Machine Learning & Evaluation (1.5)	
6	Dashboard, Insights & Decisions (2.0)	
Project Report & Documentation (1.0)		
8	Demonstration & GitHub Submission (0.5)	
	Total: 10 points	

Team Size: 1–4 students

This final project (50% of the course grade) assesses students' ability to develop an end-to-end, evidence-based decision support solution using MySQL, KNIME, a machine learning model, and an interactive dashboard.

Required pipeline: Dataset → MySQL → KNIME (data quality, preparation, analysis and ML) → MySQL (results) → Interactive Dashboard → Decision Recommendations.

Project Aims

- Define real-world problems through measurable decision questions.
- Assess data quality, build a reproducible data pipeline, and apply suitable analytical or machine learning methods.
- Translate reliable findings into practical, evidence-supported decisions.
- Communicate results in a professional written report, interactive dashboard, and demonstration.

Expected Project Outcomes

- Select and assess a relevant dataset, identifying quality risks and documenting cleaning decisions.
- Implement MySQL queries, and connect KNIME to MySQL for data retrieval and result write-back.
- Apply and evaluate at least one model covered in class: Linear Regression, Logistic Regression, K-Means, or Naive Bayes.
- Build an interactive dashboard that visualises KPIs and relevant model outputs.
- Justify at least three actionable recommendations using data and model evidence.

Project Deliverable Outcomes

Each team must complete ALL the following components. Scores total 10 points.

1. Problem Definition & Decision Questions (1.0 point)

- Describe the real business, educational, or organisational problem and identify the decision-maker.
- State the project objective and at least THREE specific, measurable decision questions.
- Explain how the planned analyses and dashboard address those questions.

2. Dataset Selection & Data Quality Assessment (1.5 points)

- Use a real, ethically sourced dataset with documented source, ownership/licence, number of records, attributes, meanings and relevance to the decision problem.
- Recommended dataset size: at least 1,000 records and five meaningful attributes. Smaller datasets may be used when justified by the problem and approved by the instructor; size alone does not imply quality.
- Conduct at least FOUR relevant quality checks covering, as applicable: completeness (missing values), uniqueness (duplicates), validity (types, formats, range rules), consistency (cross-field/table conflicts), and outliers or plausible accuracy issues.
- Summarise data quality BEFORE and AFTER cleaning, preferably in a comparison table with counts or percentages. Document transformations, removed/imputed records, and the rationale for retaining or handling outliers.
- Carry out cleaning in a reproducible KNIME workflow and discuss any remaining bias, sampling limitations, class imbalance, or other risks for model reliability.

3. MySQL Database & SQL Queries (1.0 point)

- Create and populate an appropriate MySQL database with meaningful table names, types, keys, and relationships where applicable.
- Demonstrate SQL filtering and aggregation/KPIs; use joins where applicable. Include runnable database schema, import, and query scripts.
- Document the database structure and ensure KNIME can read from MySQL.

4. KNIME Workflow & Data Processing (1.5 points)

- Use KNIME database connectors to read directly from MySQL, assess data quality, clean/transform the data and perform EDA.
- Organise the workflow into clearly labelled sections; make operations reproducible and include basic validation checks.
- Write relevant cleaned summaries and/or model predictions, scores or cluster labels BACK to one or more MySQL tables. A CSV-only workflow is not sufficient.

5. Machine Learning & Model Evaluation (1.5 points)

- Implement at least ONE model studied in class: Linear Regression, Logistic Regression, K-Means, or Naive Bayes. More models are optional.
- Explain the task, feature/target selection (for supervised learning), training/validation approach, and why the model serves a decision question.
- Use suitable evaluation: MAE/RMSE/R^2 for regression; confusion matrix and accuracy/precision/recall/F1 for classification; cluster profiling and a suitable score such as silhouette for K-Means.

- Discuss limitations, potential leakage and impacts of data quality. Interpret findings rather than reporting only model scores.

6. Dashboard, Insights & Decision Recommendations (2.0 points)

- Build an INTERACTIVE dashboard using data from MySQL, including relevant KNIME-written results. Power BI is recommended; equivalent tools are acceptable.
- Include clear KPIs, appropriate chart types, filters, and views showing descriptive findings plus predictions or segment results where relevant.
- Answer the decision questions and propose at least THREE practical recommendations. For each, specify the action, supporting evidence, and expected benefit or trade-off.
- Discuss uncertainty; avoid claims that exceed the evidence.

7. Project Report & Documentation (1.0 point)

Submit an English-language PROJECT REPORT (recommended 8–12 A4 pages, excluding references and appendices). Use labelled figures/tables, clear citations, and the following structure:

- 1. Introduction & Problem Definition — decision context, objectives, and decision questions.
- 2. Dataset Description — source, size, fields, permissions, and relevance.
- 3. Data Quality Assessment — checks, BEFORE/AFTER metrics, cleaning rationale, and remaining limitations.
- 4. System Architecture & Data Pipeline — MySQL → KNIME → model → MySQL → dashboard, with a workflow diagram.
- 5. Exploratory Data Analysis — statistics, KPIs, patterns, and charts.
- 6. Machine Learning — selected algorithm, design, evaluation metrics, interpretation, and limitations.
- 7. Dashboard & Insights — dashboard screenshots and explanations linked to questions.
- 8. Decision Recommendations — at least three evidence-based decisions with justification.
- 9. Conclusion & Limitations — findings, constraints, and potential improvements.
- References & Appendices — dataset sources, technical details, SQL samples, and KNIME screenshots as needed.

8. Demonstration & GitHub / Workspace Submission (0.5 point)

- Provide a GitHub or shared workspace link containing the PDF report, SQL scripts, shareable data or source reference, exported KNIME workflow, dashboard file, and README.
- Include instructions sufficient for a reviewer to reproduce the pipeline; document connections using placeholders, not actual passwords.

- Demonstrate the end-to-end flow: MySQL → KNIME → model/result write-back to MySQL → dashboard → recommended decisions. Every team member should be prepared to explain their contribution.

Submission Checklist

- Project report in PDF format (8–12 pages recommended, English).
- SQL scripts and schema or safe database export; data and source/licence information.
- KNIME workflow export containing quality checks, preprocessing, model and MySQL write-back.
- Interactive dashboard source file or accessible project (Power BI or equivalent).
- GitHub/workspace URL and README with setup, execution, and dependency instructions.
- Demonstration of the complete system and supporting evidence.

Suggested Topics

- Retail sales and demand forecasting — Linear Regression — inventory planning.
- Customer churn or student success risk — Logistic Regression — intervention prioritisation.
- Customer or product segmentation — K-Means — tailored marketing/resource allocation.
- Customer feedback classification — Naive Bayes — service-improvement priorities.

Important Notes & Academic Integrity

- A successful project must make meaningful decisions; the accuracy of an ML model is not the sole goal.
- Data quality assessment, a working MySQL–KNIME connection, MySQL result write-back, an interactive dashboard, and the final report are mandatory.
- Use public, anonymised, or properly authorised datasets; never expose sensitive data or database credentials.
- Reference datasets, papers, software and any AI-assisted work in accordance with the VNUK academic integrity policy on the cover page.